

# Muhan Lin

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## Education

### Purdue University

Expected: May 2029

*Ph.D. in Computer Science*

### Carnegie Mellon University

Aug 2023 – May 2025

*M.S. in Robotics*

GPA: 4.05/4.0

- **Courses:** Intro to RL (A), Multi-robot Planning (A), Mobile Manipulation (A), Advanced Computer Vision (A).

### The Chinese University of Hong Kong (Shenzhen)

Sep 2019 – May 2023

*B.Eng. in Electronic Info Engineering (Computer Engineering), First-Class Honors*

GPA: 3.6/4.0

- **Core courses:** ML (A), Algorithm Design (A), C++ (A), Python (A), Linear Algebra (A), Mechanics (A).
- **Graduate course:** Programming for Robotics (A-).

## Awards

2025 Toyota Research Institute (North America) PhD Fellowship

2025 Herbold Scholarship (Top 1.5%), Department of Computer Science, Purdue University

2019-2022 Dean's List, School of Science and Engineering, CUHK-Shenzhen

2019-2022 Bowen Scholarship, CUHK-Shenzhen

2020 3rd National Prize (Top 10%), The RISC-V Challenge of China College IC Smart-car Competition

## Selected Publications

RM-Drive: LLM-driven Reward Machines for Condition-Aware Adversarial Driving Scenario Generation

**Muhan Lin**, Kai-Chun Chang, Hideki Okamoto, Bardh Hoxha, Joseph Campbell, Georgios Fainekos  
ICRA, US Patent, In submission

MAPL: Multi-Objective Preference Learning for Robot Locomotion

Xiyue Chen, **Muhan Lin**, Shuyang Shi, Joseph Campbell  
In submission

Speaking the Language of Teamwork: LLM-Guided Credit Assignment in Multi-Agent Reinforcement Learning

**Muhan Lin**, Shuyang Shi, Vinitha Marupeddi, Yue Guo, Vaishnav Tadiparthi, Behdad Chalaki, Ehsan Moradi Pari, Simon Stepputtis, Woojun Kim, Joseph Campbell, Katia Sycara  
Neurips ARLET 2025

Navigating Noisy Feedback: Enhancing Reinforcement Learning with Error-Prone Language Models

**Muhan Lin**, Shuyang Shi, Yue Guo, Behdad Chalaki, Vaishnav Tadiparthi, Ehsan Moradi Pari, Simon Stepputtis, Joseph Campbell, Katia Sycara  
Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024

Speedup Techniques for Switchable Temporal Plan Graph Optimization

He Jiang, **Muhan Lin**, Jiaoyang Li  
(Oral presentation) Association for the Advancement of Artificial Intelligence (AAAI) 2025

Asymptotically Efficient Estimators for Range-based Robot Relative Localization

Yue Wang, **Muhan Lin**, Xinyi Xie, Yuan Gao, Fuqin Deng, Tin Lun Lam  
IEEE/ASME Transactions on Mechatronics (TMECH) 2023

## Research Experience

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**Research Resident**, Toyota Research Institute (North America) – Ann Arbor, US

### **Adversarial Traffic Generation for Safety Tests of Autonomous Driving Systems**

**Advisor:** Dr. Georgios Fainekos, Dr. Bardh Hoxha, Hideki Okamoto

May 2026 – Aug 2026

ICRA, US Patent

- Developed RL-based adversarial driving scenario generation using context-aware reward modeling.
- Built LLM-driven Reward Machines for interpretable adversarial traffic generation, automating reward designs.

**Student Researcher**, Purdue – West Lafayette, US

### **Reward Design for Robot Learning**

**Advisor:** Prof. Joseph Campbell | Purdue

Sept 2025 – Present

- Data-efficient sim-to-real robot learning from rewards inferred from one-shot internet video demonstrations.
- Learning multi-objective rewards from LLM preferences for sim-to-real quadruped locomotion.

**Student Researcher**, Robotics Institute (RI), CMU – Pittsburgh, US

### **Project 1: Enhancing Reinforcement Learning with Error-Prone Language Models**

**Advisor:** Prof. Katia Sycara | RI, CMU

Sept 2023 – Present

Neurips ARLET, EMNLP

- Extended *RL from LLM Feedback* to handle small language models prone to incorrect rankings: proposed a potential-based reward specification method to scale down rewards based on LLM feedback uncertainty.
- Theoretically and empirically showed that uncertain LLM feedback leads to uninformative potential-based rewards, avoiding misleading rewards and enabling efficient training even with small error-prone LLMs.
- Extended the proposed method to multiagent collaboration tasks for credit assignment, decomposing team rewards into individual ones with LLM feedback.

### **Project 2: Plan-Graph Optimization Acceleration for Unexpected Delays in Execution of Multi-agent Path Plans**

**Advisor:** Prof. Jiaoyang Li | RI, CMU

Feb – Aug 2024

AAAI, Course Research Project

- Proposed an accelerated optimization algorithm for the Switchable Temporal Plan Graph (STPG) to address unexpected delays when executing a multiagent path plan with techniques like incremental searching.
- Achieved over 100% improvement in STPG reconstruction success rates within 0.5 seconds across maps with diverse agent densities, outperforming *Mixed-Integer Linear Programming* and *naive graph-based methods*.
- Developed a C++-based real-time visualization tool to display updated STPGs during the replanning process.

### **Project 3: RL-based Mobile Robot Painting on Large Vertical Canvas**

**Advisor:** Prof. Zackory Erickson | RI, CMU

Feb – Apr 2024

Course Research Project

- Extended FRIDA—an algorithm controlling a robot with a fixed base to paint on a desktop small canvas—for a large canvas and a mobile manipulator based on efficient RL-driven stroke planning in real robot experiments.
- Coordinated the motion of the base and arm to move the end effector to a given stroke position by decomposing end-effector motion in horizontal and vertical directions.
- Utilized AR tags to realize coordinate system transformation between the robot and canvas.

### **Project 4: Active Feature Selection for Pose Estimation of Multi-spectral Odometry**

**Advisor:** Prof. Sebastian Scherer | RI, CMU

May - Nov 2022

RISS Paper

- Developed a quality measurement metric to actively select visual feature points for the visual odometry.
- Improved the accuracy and efficiency of visual odometry pose estimation (13.036% of the original residuals, 58.832% of the original CPU usage) with filtered visual feature points in dark environments.
- Studied the multi-camera data fusion scheme for visual odometry.
- Integrated the active feature selection method into a multi-camera multi-spectral odometry system with C++ to assist multi-sensor data fusion and pose estimation in challenging environments with fewer features.

**Research Assistant**, CUHK-Shenzhen & Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS) – China

### **Project 1: Initial Relative Pose Estimation with the Two-stage Extended Kalman Filter**

**Advisor:** Prof. Tin Lun Lam | School of Science and Engineering, CUHK-Shenzhen

Jan - May 2022

*Bachelor Thesis*

- Developed a Two-stage Extended Kalman Filter for improved *robot-to-robot* initial relative pose estimation using multi-sensor data fusion, improving LIDAR map merging accuracy.
- Validated the method in simulated environments using ROS and Gazebo.

### **Project 2: Multi-robot Relative Pose Estimation with UWB and Odometry Information**

**Advisor:** Prof. Tin Lun Lam | Director of Center for Intelligent Robots, AIRS

Feb 2021 – Jan 2022

*TMECH Paper*

- Enhanced localization accuracy in multiagent systems, where robots communicated their odometry trajectories and relative distance with each other and then estimated their initial relative poses.
- Developed a real-time relative-pose estimation system structure compatible with multiple offline initial pose estimation algorithms, which adopt the particle filter and the Kalman filter to track subsequent poses.
- Led the design, assigned tasks to younger teammates, and performed the physical experiments of a novel relative localization algorithm with ROS, MATLAB, and a motion-capturing system.

## **Competition**

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### **The Smart Car and Real-time OS Application Development**

Mar – Aug 2020

*The RISC-V Challenge of China College IC Smart-car Competition*

Shenzhen & Nanjing, China

**Advisor:** Prof. Simon Pun | CUHKSZ-Stanford Maker Lab

**3rd National Prize (Top 10%)**

- Programmed registers on a bare MCU and read and processed raw sensor signals to control a smart car with a linear CCD in C/C++.
- Applied a real-time operating system to the smart car to support multi-threading work to improve real-time performance and resource utilization efficiency.
- Designed a novel path-tracking algorithm in C for the paths bounded by dashed lines with infrared sensors. Increased the success rate of the original algorithm from approximately 50% to above 95%.

## **Academic Activities**

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**Reviewer:** RAL 2025, Neurips ARLET 2025, ICRA 2024 - 2026 and IEEE Transactions on Games 2024

## **Skills**

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**Programming:** Python, C/C++, MATLAB, Verilog, VHDL, Mysql, Go, HTML, CSS

**Software & Tools:** ROS, PyTorch, RLlib, TorchRL, OpenAI Gym, Moveit, OpenCV, Django, FreeRTOS